



GaugeHD Synthetic — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 29-March-2026

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1 Test Results — GaugeHD Synthetic (MKM-GHD-001)

Tests run on **29 March 2026 at 10:29:14.**

Table 1: Test Results Summary — GaugeHD Synthetic (MKM-GHD-001)

Total Tests	62
Passed	62
Failed	0
Skipped	0
Duration	9.40s

Table 2: Detailed Test Results — GaugeHD Synthetic

Test Description	Acceptance Criteria	Result	Time
<code>_snap_to_river</code> skips zero-length segments without error.	<code>isinstance(lat, float); isinstance(lon, float)</code>	Pass	0.000s
<code>_create_synthetic_gauge</code> returns None when flanking gauge not in lookup.	<code>result is None</code>	Pass	0.003s
Lines 83-110: <code>generate_all_gauge_histories</code> processes all gauges.	<code>isinstance(result, list); len(result) == 1</code>	Pass	0.217s
Lines 105-107: error on one gauge -j logged, continues.	<code>isinstance(result, list)</code>	Pass	0.356s
All levels positive	<code>all(obs["level_meters"] > 0 for obs in result)</code>	Pass	0.006s
Correct length approximately	<code>728 <= len(result) <= 732</code>	Pass	0.003s
Dates are sequential	<code>(d_curr - d_prev).days == 1</code>	Pass	0.004s
Different seeds different output	<code>levels1 != levels2</code>	Pass	0.012s
Each obs has date and level	<code>"date" in obs; "level_meters" in obs (+1 more)</code>	Pass	0.001s
Gauge data without FloodGauge wrapper uses defaults.	<code>isinstance(result, list)</code>	Pass	0.001s
When historical high date is within range, that day gets historical high level.	<code>match[0]["level_meters"] == 7.5</code>	Pass	0.001s
Invalid date format doesn't crash.	<code>isinstance(result, list)</code>	Pass	0.001s
Historical high date before <code>start_date</code> is ignored.	<code>isinstance(result, list)</code>	Pass	0.001s

Test Description	Acceptance Criteria	Result	Time
Lines 176-179: Historical high injected via fallback when date not matched in loop.	<code>len(match) == 1;</code> <code>match[0]["level_meters"] == 8.0</code>	Pass	0.001s
Gauge data with no FloodStages uses sensible defaults.	<code>isinstance(result, list); len(result) > 0</code>	Pass	0.001s
Lines 97-100: Partially tidal gauge uses dampened tidal signal.	<code>len(result) > 700;</code> <code>all(lev > 0 for lev in levels)</code>	Pass	0.003s
Returns list	<code>isinstance(result, list)</code>	Pass	0.001s
Seed reproducible	<code>r1 == r2</code>	Pass	0.003s
Lines 92-95: Tidal gauge uses wider daily_std, tidal_amplitude > 0.	<code>len(result) > 700;</code> <code>max(levels) > min(levels)</code>	Pass	0.003s
Lines 147-149: Tidal component with spring-neap cycle.	<code>11 != 12</code>	Pass	0.003s
Lines 138-141: historical high injected after main loop.	<code>len(match) == 1;</code> <code>match[0]["level_meters"] == 9.99</code>	Pass	0.004s
When historical_high_dt is None (out of range), post-loop block skipped.	<code>all(o["level_meters"] != 8.0 for o in result)</code>	Pass	0.001s
When historical_high_date is None, both branches skipped.	<code>isinstance(result, list); len(result) > 0</code>	Pass	0.001s
generate() returns count=0 when GAUGE_POINTS has > 2 entries.	<code>result["count"] == 0</code>	Pass	0.003s
generate() returns count=0 when GAUGE_POINTS is None.	<code>result["count"] == 0</code>	Pass	0.001s
Lines 74-75: missing file raises FileNotFoundError.	—	Pass	0.000s
Lines 72-80: load_gauge_portfolio reads gauge.json.	<code>isinstance(result, list); len(result) == 1</code>	Pass	0.003s
_load_river_polyline returns None when cache file doesn't exist.	<code>result is None or isinstance(result, list)</code>	Pass	0.001s
_snap_to_river returns original coords when river is None.	<code>lat == 51.46; lon == -0.30</code>	Pass	0.000s
Lines 143-144: bad CSV -i error logged, continues.	<code>isinstance(result, list)</code>	Pass	0.002s
Lines 113-147: process_nra_directory scans CSV files.	<code>isinstance(result, list); len(result) == 1</code>	Pass	0.002s
Line 134: output_dir=None uses config.get_gaugehd_dir().	<code>isinstance(result, list)</code>	Pass	0.002s
Lines 131-147: no matching files -i empty list.	<code>result == []</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Property projecting to seg_idx=0, t~0 should not create a gauge.	result["count"] == 0	Pass	0.002s
Propertyhc has synth in nearest gauges	found_synth	Pass	0.011s
Synthetic gauge in propertyhc should have basis_bps computed.	"basis_bps" in ng	Pass	0.007s
Every property has one synthetic gauge	not failures	Pass	2.255s
Every property has three gauges total	len(ngs) == 3	Pass	0.160s
Every property has two real gauges	not failures	Pass	1.938s
Properties should have flood_events sourced from gauge storm responses.	properties_with_events > 0	Pass	1.952s
At least some properties should have flooded=True events.	flooded.count > 0	Pass	1.951s
Appends to gauge json	len(data["flood_gauges"]) > 3; len(synth) > 0	Pass	0.008s
Creates synthetic gauges	result["count"] > 0; len(result["gauge_ids"]) == result["count"]	Pass	0.009s
Two properties near the same river point should share a synthetic gauge.	result["count"] == 2	Pass	0.017s
Property far beyond polyline end should not create a synthetic gauge.	result["count"] == 0	Pass	0.002s
Interpolated elevation	3.0 <= elev <= 5.0	Pass	0.009s
Interpolated flood stages	3.5 <= stages["FloodAlert"] <= 4.5; 4.5 <= stages["FloodWarning"] <= 5.5 (+1 more)	Pass	0.013s
No gauge json returns zero	result["count"] == 0	Pass	0.000s
No property json returns zero	result["count"] == 0	Pass	0.006s
Synthetic gauge has cdm structure	"GaugeID" in fg["Header"]; "CatchmentID" in fg["Header"] (+8 more)	Pass	0.009s
Synthetic gauge lat/lon should be on the polyline (very close to it).	dist_m < 50.0; dist_m < 1.0	Pass	0.014s
A pre-existing SYNTH- gauge in gauge.json is skipped in polyline matching.	result["count"] >= 0	Pass	0.007s
Gauge json has synth gauges	len(synth) > 0	Pass	0.002s

Test Description	Acceptance Criteria	Result	Time
Synth gauge has required cdm fields	<code>fg["Header"]["GaugeID"].startswith("SYNTH")</code> <code>fg["Location"]["GaugeLatitude"] != 0</code> (+7 more)	Pass	0.001s
Gaugehc has synth curves	<code>len(synth) > 0</code>	Pass	0.014s
Synth has term structure	<code>len(ts) > 0</code>	Pass	0.007s
Synth hazard rates positive	<code>hc.get("annual.hazard.rate.alert", 0)</code> <code>> 0</code>	Pass	0.008s
Synth gaugets files exist	<code>len(synth_files) > 0</code>	Pass	0.000s
At least some storms should exceed alert at synthetic gauges.	<code>len(exceeded) > 0</code>	Pass	0.196s
Synth gaugets have storm responses	<code>len(resps) > 0</code>	Pass	0.150s
Property with lat=0 is silently skipped.	<code>result["count"] == 0</code>	Pass	0.005s
Property with lon=0 is silently skipped.	<code>result["count"] == 0</code>	Pass	0.002s

1.1 Test Document History

Date	Version	Description	Author
27-Mar-2026	1.0	Initial test suite (80 tests)	D. Kelly
29-Mar-2026	1.1	18 test(s) removed (total: 62)	D. Kelly